2026 AMD AI DEVMASTER · TRACK 2

PrivateOps Agent

A permission-first private computer AI on AMD Radeon GPU + ROCm.

BROAD VISIBILITY	DELIBERATE ACTION	EVIDENCE FOR DECISIONS
Supported observation	Exact-action confirmation	Local audit + Radeon proof

Version 1.0 · English submission package · MIT License

01

Application scenario

People increasingly say, 'I gave my whole computer to my AI.' Useful AI needs permission if it is going to take work away from people—but permission must not mean silent control.

PRODUCT PROMISE	Broad visibility. Deliberate action. Evidence for every decision. PrivateOps observes supported computer and project state, defaults to read-only behavior, and requires a second exact-action confirmation before any supported side effect.

Two truthful surfaces

Windows exposes seven fixed read-only system collectors. Radeon Cloud exposes private project and runtime observation beside the local model. Whole-computer permission means broad supported visibility, never unrestricted administrator access.

Independent developers	Computer observation, release audits, competition packaging, and reproducible checks.
Small private teams	Local knowledge operations where source, credentials, and internal plans cannot leave the controlled environment.

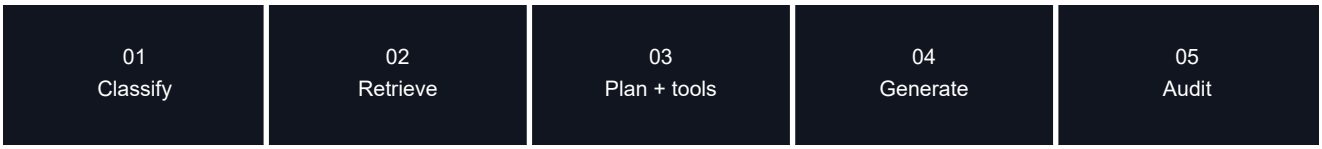
Distinctive value

- Safe, confirm, and blocked decisions remain reviewer-visible.
- One-use tokens bind permission to one exact action.
- All five Track 2 capability categories reinforce one permission-first workflow.

02

Architecture and data flow

A dependency-light Python service orchestrates local evidence, planning, guarded tools, memory, model inference, audit, and benchmark proof. A no-build browser console exposes every layer.



Component	Responsibility	Trust property
Retrieval	Indexes approved text files and creates stable evidence IDs.	Answers expose source and snippet.
Permission	Classifies requests as safe, confirm, or blocked.	Side effects never execute silently.
Tool registry	Inventory, search, tests, and readiness checks.	Paths stay confined; commands are allow-listed.
Planner	Creates retrieve, inspect, and synthesize steps.	Status and results remain visible.
Local state	Memory, JSONL audit, profile, and benchmarks.	Atomic writes and credential redaction.
ROCm provider	Calls a loopback OpenAI-compatible server.	No closed remote core dependency.

Failure behavior

Malformed and oversized requests receive stable errors. Non-loopback competition endpoints, path escape, shell metacharacters, expired or reused confirmation tokens, and unsupported verified benchmark claims are rejected. ROCm mode never silently falls back to development output.

03

Core capabilities

Local RAG	Fourteen bilingual sources, Chinese bigrams, stable IDs, bounded snippets.	**Tool calling**	Project tools plus seven fixed Windows collectors with structured results.
Multi-step planning	Inspectable retrieval, tool work, and synthesis stages.	**Local memory**	Project facts and preferences in atomic local JSON state.
Privacy controls	Three risk levels, one-use tokens, path confinement, and allow-listing.	**Audit ledger**	Chronological records of evidence, tools, provider, latency, and memory.

Agent framework

PrivateOps uses LangChain Core StructuredTool adapters as its open Agent tool framework. Each adapter delegates to the permission-aware registry, preserving path confinement, confirmation tokens, allow-listing, structured results, and audit recording.

Interaction design

The Local Trust Console keeps proof adjacent to the user task and exposes an explicit Remote model / Local ROCm selector. Remote mode requires its own API key; Local ROCm restores the loopback Qwen defaults and accepts an empty key. The interface is responsive, keyboard-focusable, and honors reduced-motion preferences.

04

Model and local deployment

Competition mode runs an open-weight instruct model through vLLM on a loopback endpoint inside the same Radeon Cloud instance as PrivateOps.

Mode	Purpose	Runtime contract
Remote model	Optional integration testing on ordinary computers.	HTTP(S) endpoint and key required.
Local ROCm	ROCm vLLM on 127.0.0.1 in the same instance.	Loopback required; key optional.

Default profile

Qwen/Qwen2.5-7B-Instruct is the shared competition default. vLLM serves it on 127.0.0.1:8000 with an 8,192-token context and 0.85 GPU-memory utilization. The Apache-2.0 model remains configurable when the cloud image requires another compatible open artifact.

Privacy boundary

- The browser talks only to the PrivateOps service.
- The service retrieves files only from the configured workspace.
- The service calls the model on a validated loopback URL.
- API keys are omitted from status, audit, and browser responses.
- Core behavior does not call shared public model APIs.

Reproducibility

Separate commands start the ROCm model server, application, benchmark collector, tests, smoke check, and final material validator. The cloud runbook includes rocminfo, HIP, PyTorch, endpoint, tunnel, and evidence commands.

05

Radeon optimization and measurement

A verified same-instance run captured AMD Radeon Graphics (gfx1100), ROCm 7.2.1, ROCm vLLM, and Qwen2.5-7B-Instruct. Final metrics are generated from the immutable machine record.

Implemented optimization

- 8,192-token context and 0.85 GPU-memory utilization preserve headroom.
- Loopback transport keeps prompts local and avoids public-network latency.
- Top-k RAG, environment exclusion, bounded samples, and a 25-item list limit reduce context.
- Temperature zero improves repeatability during evidence collection.

Metric	Verified value	Scope
Direct latency	1,511.072 ms	Loopback completion.
Throughput	32.427 tokens/s	49 completion tokens.
Complete Agent	18,384.639 ms	RAG, tool, model, audit.

INTEGRITY RULE No baseline speedup, P50/P95, VRAM peak, concurrency, or hardware soak value is claimed without a new Radeon measurement.

06

Verification and deliverables

Automated coverage

The 102-test local finalization baseline covers bilingual retrieval, explicit remote/local provider validation, ROCm profile, LangChain tools, bounded inventories, permission and confirmation, memory, audit integrity, benchmark provenance, Agent orchestration, HTTP errors, Markdown safety, UI state, the provider-required smoke path, and unresolved submission placeholders. A repeatable software stability check verifies visible provider failure and zero silent fallback.

Reviewer deliverables

- Complete source, tests, local and cloud startup scripts.
- Project specification PDF and editable HTML source.
- Editable English competition PPT with source notes.
- Architecture SVG/PNG and one-page poster PDF.
- Timed demo narration and exact shot list.
- Radeon Cloud runbook and benchmark evidence schema.
- English PR copy and final checklist.

Completed proof and public submission

The real Radeon Cloud run, measured evidence, public demonstration video, official repository Fork, and Pull Request #87 are complete. The video is available at https://www.bilibili.com/video/BV1nQuA6qEFS/ and the Pull Request is available at https://github.com/AMD-DEV-CONTEST/Radeon-hackathon-2026-07/pull/87.

English-subtitled version: https://www.bilibili.com/video/BV1A1M66VEqS/.

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